

Limits of Rational Functions

A. If $f(x)$ is a rational function given by $f(x) = \frac{p(x)}{q(x)}$, such that $p(x)$ and $q(x)$ have no common factors, and c is a real number such that $q(c) = 0$, then

I. $\lim_{x \rightarrow c} f(x)$ does not exist

II. $\lim_{x \rightarrow c} f(x) = \pm\infty \longrightarrow x = c$ is a vertical asymptote

B. If $f(x)$ is a rational function given by $f(x) = \frac{p(x)}{q(x)}$, such that reducing a common factor between $p(x)$ and $q(x)$ results in the agreeable function $k(x)$, then

$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \frac{p(x)}{q(x)} = \lim_{x \rightarrow c} k(x) = k(c) \longrightarrow$ Hole at the point $(c, k(c))$

Limits of a Function as x Approaches Infinity

If $f(x)$ is a rational function given by $f(x) = \frac{p(x)}{q(x)}$, such that $p(x)$ and $q(x)$ are both polynomial functions, then

A. If the degree of $p(x) > q(x)$, $\lim_{x \rightarrow \infty} f(x) = \infty$

B. If the degree of $p(x) < q(x)$, $\lim_{x \rightarrow \infty} f(x) = 0 \longrightarrow y = 0$ is a horizontal asymptote

C. If the degree of $p(x) = q(x)$, $\lim_{x \rightarrow \infty} f(x) = c$, where c is the ratio of the leading coefficients.
 $\longrightarrow y = c$ is a horizontal asymptote

Special Trig Limits

A. $\lim_{x \rightarrow 0} \frac{\sin ax}{ax} = 1$

B. $\lim_{x \rightarrow 0} \frac{ax}{\sin ax} = 1$

C. $\lim_{x \rightarrow 0} \frac{1 - \cos ax}{ax} = 0$

L'Hospital's Rule

If results $\lim_{x \rightarrow c} f(x)$ or $\lim_{x \rightarrow \infty} f(x)$ results in an indeterminate form $\left(\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \cdot \infty, 0^0, 1^\infty, \infty^0\right)$, and $f(x) = \frac{p(x)}{q(x)}$, then

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \frac{p(x)}{q(x)} = \lim_{x \rightarrow c} \frac{p'(x)}{q'(x)} \quad \text{and} \quad \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} = \lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)}$$

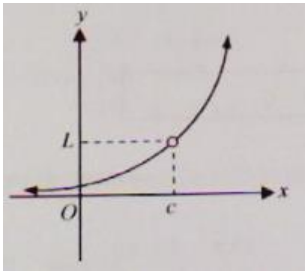
The Definition of Continuity

A function $f(x)$ is continuous at c if

- I. $\lim_{x \rightarrow c} f(x)$ exists
- II. $f(c)$ exists
- III. $\lim_{x \rightarrow c} f(x) = f(c)$

Types of Discontinuities

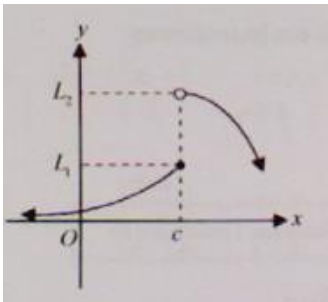
Removable Discontinuities (Holes)



- I. $\lim_{x \rightarrow c} f(x) = L$ (the limit exists)
- II. $f(c)$ is undefined

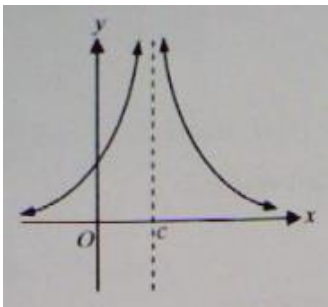
Non-Removable Discontinuities (Jumps and Asymptotes)

A. Jumps



$$\lim_{x \rightarrow c} f(x) = DNE \text{ because } \lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x)$$

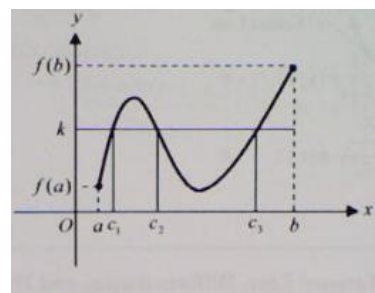
B. Asymptotes (Infinite Discontinuities)



$$\lim_{x \rightarrow c} f(x) = \pm \infty$$

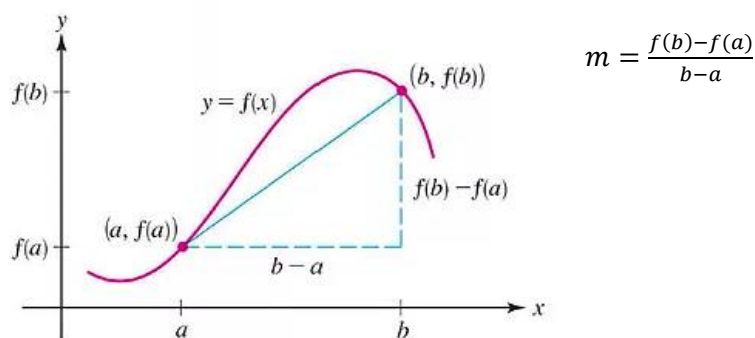
Intermediate Value Theorem

If f is a continuous function on the closed interval $[a, b]$ and k is any number between $f(a)$ and $f(b)$, then there exists at least one value of c on $[a, b]$ such that $f(c) = k$. In other words, on a continuous function, if $f(a) < f(b)$, any y -value greater than $f(a)$ and less than $f(b)$ is guaranteed to exist on the function f .



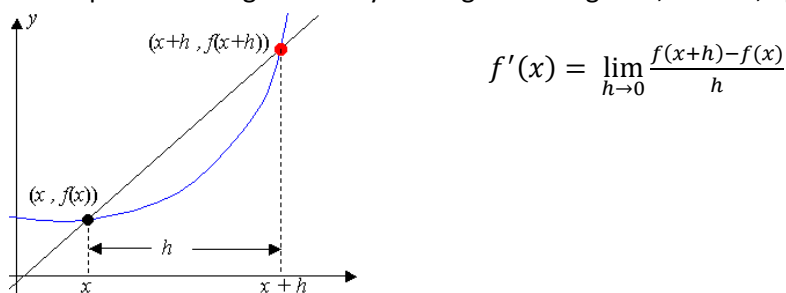
Average Rate of Change

The average rate of change, m , of a function f on the interval $[a, b]$ is given by the slope of the secant line.

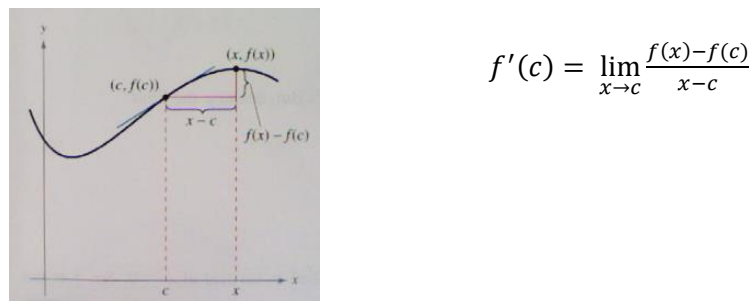


Definition of the Derivative

The derivative of the function f , or instantaneous rate of change, is given by converting the slope of the secant line to the slope of the tangent line by making the change in x , Δx or h , approach zero.



Alternate Definition



Differentiability and Continuity Properties

- A. If $f(x)$ is differentiable at $x = c$, then $f(x)$ is continuous at $x = c$.
- B. If $f(x)$ is not continuous at $x = c$, then $f(x)$ is not differentiable at $x = c$.
- C. The graph of f is continuous, but not differentiable at $x = c$ if:
- I. The graph has a cusp or sharp point at $x = c$
 - II. The graph has a vertical tangent line at $x = c$
 - III. The graph has an endpoint at $x = c$

Basic Derivative Rules

Given c is a constant,

1. Constant Rule $\frac{d}{dx} [c] = 0$
2. Constant Multiple Rule $\frac{d}{dx} [cf(x)] = cf'(x)$
3. Sum Rule $\frac{d}{dx} [f(x) + g(x)] = f'(x) + g'(x)$
4. Difference Rule $\frac{d}{dx} [f(x) - g(x)] = f'(x) - g'(x)$
5. Product Rule $\frac{d}{dx} [f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$
6. Quotient Rule $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$
7. Chain Rule $\frac{d}{dx} [f(g(x))] = f'(g(x))g'(x)$

Derivatives of Trig Functions

- | | | |
|--|---|--|
| 1. $\frac{d}{dx} [\sin x] = \cos x$ | 2. $\frac{d}{dx} [\cos x] = -\sin x$ | 3. $\frac{d}{dx} [\tan x] = \sec^2 x$ |
| 4. $\frac{d}{dx} [\sec x] = \sec x \tan x$ | 5. $\frac{d}{dx} [\csc x] = -\csc x \cot x$ | 6. $\frac{d}{dx} [\cot x] = -\csc^2 x$ |

Derivatives of Inverse Trig Functions

- | | | |
|---|--|--|
| 1. $\frac{d}{dx} [\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$ | 2. $\frac{d}{dx} [\cos^{-1} x] = \frac{-1}{\sqrt{1-x^2}}$ | 3. $\frac{d}{dx} [\tan^{-1} x] = \frac{1}{1+x^2}$ |
| 4. $\frac{d}{dx} [\sec^{-1} x] = \frac{1}{ x \sqrt{x^2-1}}$ | 5. $\frac{d}{dx} [\csc^{-1} x] = \frac{-1}{ x \sqrt{x^2-1}}$ | 6. $\frac{d}{dx} [\cot^{-1} x] = \frac{-1}{1+x^2}$ |

Derivatives of Exponential and Logarithmic Functions

$$1. \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$$

$$2. \frac{d}{dx} [\log_a x] = \frac{1}{x \ln a}, (a > 0, a \neq 1)$$

$$3. \frac{d}{dx} [\ln x] = \frac{1}{x}$$

$$4. \frac{d}{dx} [\ln |x|] = \frac{1}{x}$$

$$5. \frac{d}{dx} [\log_a |x|] = \frac{1}{x \ln a}, (a > 0, a \neq 1)$$

$$6. \frac{d}{dx} [e^x] = e^x$$

$$7. \frac{d}{dx} [a^x] = a^x \ln a$$

Explicit and Implicit Differentiation

A. Explicit Functions: Function y is written only in terms of the variable x ($y = f(x)$). Apply derivatives rules normally.

B. Implicit Differentiation: An expression representing the graph of a curve in terms of both variables x and y .

I. Differentiate both sides of the equation with respect to x . (terms with x differentiate normally, terms with y are multiplied by $\frac{dy}{dx}$ per the chain rule)

II. Group all terms with $\frac{dy}{dx}$ on one side of the equation and all other terms on the other side of the equation.

III. Factor $\frac{dy}{dx}$ and express $\frac{dy}{dx}$ in terms of x and y .

Tangent Lines and Normal Lines

A. The equation of the tangent line at a point $(a, f(a))$: $y - f(a) = f'(a)(x - a)$

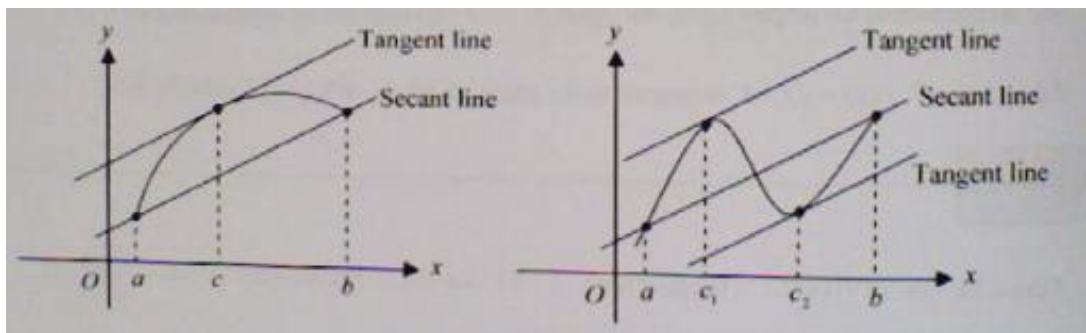
B. The equation of the normal line at a point $(a, f(a))$: $y - f(a) = -\frac{1}{f'(a)}(x - a)$

Mean Value Theorem for Derivatives

If the function f is continuous on the close interval $[a, b]$ and differentiable on the open interval (a, b) , then there exists at least one number c between a and b such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

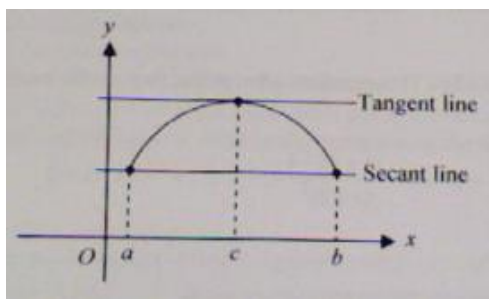
The slope of the tangent line is equal to the slope of the secant line.



Rolle's Theorem (Special Case of Mean Value Theorem)

If the function f is continuous on the close interval $[a, b]$ and differentiable on the open interval (a, b) , and $f(a) = f(b)$, then there exists at least one number c between a and b such that

$$f'(c) = \frac{f(b)-f(a)}{b-a} = 0$$



Particle Motion

A velocity function is found by taking the derivative of position. An acceleration function is found by taking the derivative of a velocity function.

$x(t)$	Position	
$x'(t) = v(t)$	Velocity	* $ v(t) = \text{speed}$
$x''(t) = v'(t) = a(t)$	Acceleration	

Rules:

- A. If velocity is positive, the particle is moving right or up. If velocity is negative, the particle is moving left or down.
- B. If velocity and acceleration have the same sign, the particle speed is increasing. If velocity and acceleration have opposite signs, speed is decreasing.
- C. If velocity is zero and the sign of velocity changes, the particle changes direction.

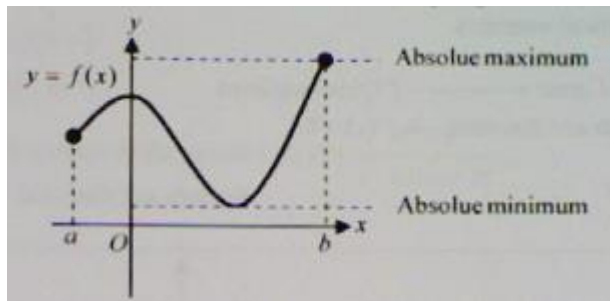
Related Rates

- A. Identify the known variables, including their rates of change and the rate of change that is to be found. Construct an equation relating the quantities whose rates of change are known and the rate of change to be found.
- B. Implicitly differentiate both sides of the equation with respect to time. (Remember: DO NOT substitute the value of a variable that changes throughout the situation before you differentiate. If the value is constant, you can substitute it into the equation to simplify the derivative calculation).
- C. Substitute the known rates of change and the known values of the variables into the equation. Then solve for the required rate of change.

*Keep in mind, the variables present can be related in different ways which often involves the use of similar geometric shapes, Pythagorean Theorem, etc.

Extrema of a Function

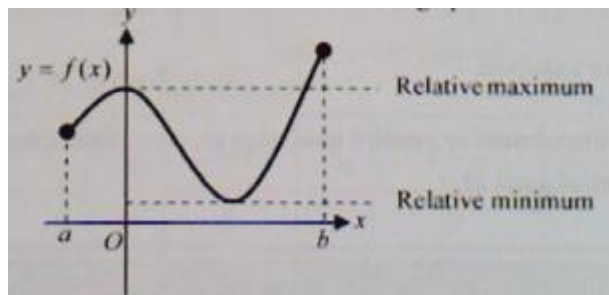
A. Absolute Extrema: An absolute maximum is the highest y – value of a function on a given interval or across the entire domain. An absolute minimum is the lowest y – value of a function on a given interval or across the entire domain.



B. Relative Extrema

I. Relative Maximum: The y -value of a function where the graph of the function changes from increasing to decreasing. Another way to define a relative maximum is the y -value where derivative of a function changes from positive to negative.

II. Relative Minimum: The y -value of a function where the graph of the function changes from decreasing to increasing. Another way to define a relative minimum is the y -value where derivative of a function changes from negative to positive.



Critical Value

When $f(c)$ is defined, if $f'(c) = 0$ or f' is undefined at $x = c$, the values of the x – coordinate at those points are called critical values.

*If $f(x)$ has a relative extrema at $x = c$, then c is a critical value of f .

Extreme Value Theorem

If the function f continuous on the closed interval $[a, b]$, then the absolute extrema of the function f on the closed interval will occur at the endpoints or critical values of f .

*After identifying critical values, create a table with endpoints and critical values. Calculate the y – value at each of these x values to identify the extrema.

Increasing and Decreasing Functions

For a differentiable function f

- A. If $f'(x) > 0$ in (a, b) , then f is increasing on (a, b) $\longrightarrow$ Tangent line has a positive slope
- B. If $f'(x) < 0$ in (a, b) , then f is decreasing on (a, b) $\longrightarrow$ Tangent line has a negative slope
- C. If $f'(x) = 0$ in (a, b) , then f is constant on (a, b) $\longrightarrow$ Tangent line has a zero slope (horizontal)

First Derivative Test

After calculating any discontinuities of a function f and calculating the critical values of a function f , create a sign chart for f' , reflecting the domain, discontinuities, and critical values of a function f .

- A. If $f'(x)$ changes sign from negative to positive at $x = c$, then $f(c)$ is a relative minimum of f .
- B. If $f'(x)$ changes sign from positive to negative at $x = c$, then $f(c)$ is a relative maximum of f .

*If there is no sign change of $f'(x)$, there exists a shelf point

Concavity

For a differentiable function $f(x)$,

- A. If $f''(x) > 0$, the graph of $f(x)$ is concave up
 $\longrightarrow$ This means $f'(x)$ is increasing
- B. If $f''(x) < 0$, the graph of $f(x)$ is concave down
 $\longrightarrow$ This means $f'(x)$ is decreasing

Second Derivative Test

For a function $f(x)$ that is continuous at $x = c$

- A. If $f'(c) = 0$ and $f''(c) > 0$, then $f(c)$ is a relative minimum.
- B. If $f'(c) = 0$ and $f''(c) < 0$, then $f(c)$ is a relative maximum.

* If $f'(c) = 0$ and $f''(c) = 0$, you must use the first derivative test to determine extrema

Point of Inflection

Let f be a function whose second derivative exists on any interval. If f is continuous at $x = c$, $f''(c) = 0$ or $f''(c)$ is undefined, and $f''(x)$ changes sign at $x = c$, then the point $(c, f(c))$ is a point of inflection.

Optimization

Finding the largest or smallest value of a function subject to some kind of constraints.

- A.** Define the primary equation for the quantity to be maximized or minimized. Define a feasible domain for the variables present in the equation.
- B.** If necessary, define a secondary equation that relates the variables present in the primary equation. Solve this equation for one of the variables and substitute into the primary equation.
- C.** Once the primary equation is represented in a single variable, take the derivative of the primary equation.
- D.** Find the critical values using the derivative calculated.
- E.** The optimal solution will more than likely be found at a critical value from **D**. Keep in mind, if the critical values do not represent a minimum or a maximum, the optimal solution may be found at an endpoint of the feasible domain.

Derivative of an Inverse

If f and its inverse g are differentiable, and the point $(c, f(c))$ exists on the function f meaning the point $(f(c), c)$ exists on the function g , then

$$\frac{d}{dx}[g(x)] = \frac{1}{f'(f^{-1}(x))} = \frac{1}{f'(f(c))}$$

BC Only: Derivatives of Parametric Functions

If f and g are continuous functions of t on an interval, then the equations $x = f(t)$ and $y = g(t)$ are called parametric equations, providing the position in the coordinate plane, and t is called the parameter.

- A.** The slope of the curve at the point (x, y) is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \text{ provided } dx/dt \neq 0$$

- B.** The second derivative at the point (x, y) is

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Antiderivatives

If $F'(x) = f(x)$ for all x , $F(x)$ is an antiderivative of f .

$$\int f(x) = F(x) + C$$

* The antiderivative is also called the Indefinite Integral

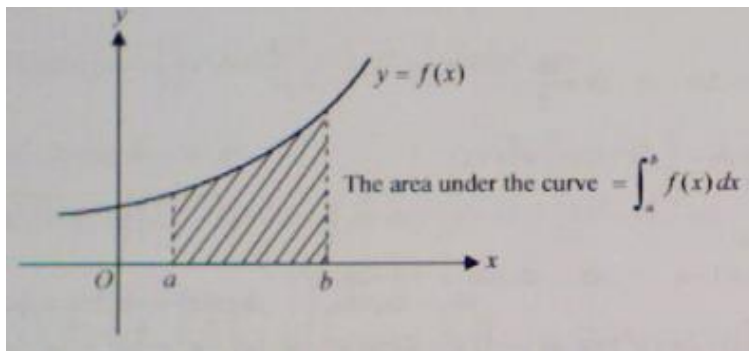
Basic Integration Rules

Let k be a constant.

$\int 0 \, dx = C$	$\int k \, dx = kx + C$
$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$	$\int \cos x \, dx = \sin x + C$
$\int \sin x \, dx = -\cos x + C$	$\int \sec^2 x \, dx = \tan x + C$
$\int \sec x \tan x \, dx = \sec x + C$	$\int \csc^2 x \, dx = -\cot x + C$
$\int \csc x \cot x \, dx = -\csc x + C$	

Definite Integrals (The Fundamental Theorem of Calculus)

A definite integral is an integral with upper and lower limits, a and b , respectively, that define a specific interval on the graph. A definite integral is used to find the area bounded by the curve and an axis on the specified interval (a, b) .



If $F(x)$ is the antiderivative of a continuous function $f(x)$, the evaluation of the definite integral to calculate the area on the specified interval (a, b) is the First Fundamental Theorem of Calculus:

$$\int_a^b f(x) dx = F(b) - F(a)$$

Integration Rules for Definite Integrals

1. $\int_a^a f(x) dx = 0$
2. $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ for $c \in (a, b)$
3. $\int_a^b f(x) dx = -\int_b^a f(x) dx$
4. $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$, where f is an **even function**.
5. $\int_{-a}^a f(x) dx = 0$, where f is an **odd function**.

*This means that c is a value of x , lying between a and b

Partial Derivatives

Partial Derivatives

Just as derivatives can be used to explore the properties of functions of 1 variable, so also derivatives can be used to explore functions of 2 variables. In this section, we begin that exploration by introducing the concept of a *partial derivative* of a function of 2 variables.

In particular, we define the *partial derivative* of $f(x, y)$ with respect to x to be

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

when the limit exists. That is, we compute the derivative of $f(x, y)$ as if x is the variable and all other variables are held constant. To facilitate the computation of partial derivatives, we define the operator

$$\frac{\partial}{\partial x} = \text{“The partial derivative with respect to } x\text{”}$$

Alternatively, other notations for the partial derivative of f with respect to x are

$$f_x(x, y) = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \partial_x f(x, y)$$

EXAMPLE 1 Evaluate f_x when $f(x, y) = x^2y + y^2$.

Solution: To do so, we write

$$f_x(x, y) = \frac{\partial}{\partial x} (x^2y + y^2) = \frac{\partial}{\partial x} x^2y + \frac{\partial}{\partial x} y^2$$

and then we evaluate the derivative as if y is a constant. In particular,

$$f_x(x, y) = y \frac{\partial}{\partial x} x^2 + \frac{\partial}{\partial x} y^2 = y \cdot 2x + 0$$

That is, y factors to the front since it is considered constant with respect to x . Likewise, y^2 is considered constant with respect to x , so that its derivative with respect to x is 0. Thus, $f_x(x, y) = 2xy$.

Likewise, the *partial derivative* of $f(x, y)$ with respect to y is defined

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$

when the limit exists. That is, we evaluate f_y as if y is varying and all other quantities are constant. Moreover, we also define the operator

$$\frac{\partial}{\partial y} = \text{"The partial derivative with respect to } y\text{"}$$

and we often use other notations for $f_y(x, y)$:

$$f_y(x, y) = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \partial_y f(x, y)$$

EXAMPLE 2 Find f_x and f_y when $f(x, y) = y \sin(xy)$

Solution: To find f_x , we use the chain rule

$$f_x = y \frac{\partial}{\partial x} \sin(xy) = y \cos(xy) \frac{\partial}{\partial x} xy = y^2 \cos(xy)$$

However, to find f_y , we begin with the product rule:

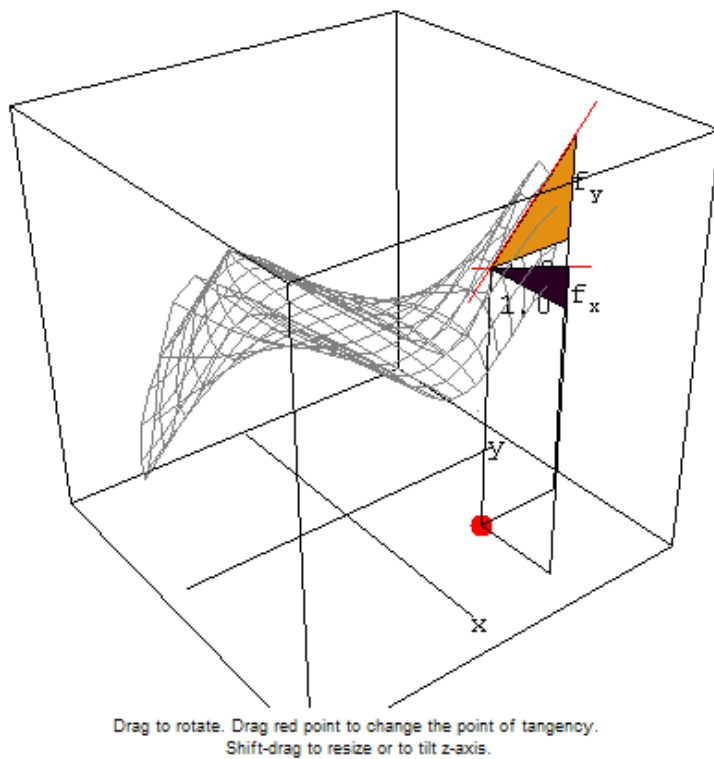
$$f_y = \frac{\partial}{\partial y} [y \sin(xy)] = \left(\frac{\partial}{\partial y} y \right) \sin(xy) + y \frac{\partial}{\partial y} \sin(xy)$$

We then use the chain rule to evaluate $\partial_y \sin(xy)$:

$$\begin{aligned} f_y &= \sin(xy) + y \cos(xy) \frac{\partial}{\partial y} (xy) \\ &= \sin(xy) + xy \cos(xy) \end{aligned}$$

If $y = q$ for some constant q , then $f(x, q)$ is a function of x , and $f_x(x, q)$ is the slope of the tangent line to the curve $z = f(x, q)$ in the $y = q$ plane. Similarly, $f_y(p, y)$ for p constant is the slope of a tangent line to the curve $z = f(p, y)$ in

the $x = p$ plane.



That is, $f_x(x, y)$ is the slope of a tangent line to $z = f(x, y)$ parallel to the xz -plane, while $f_y(x, y)$ is the slope of a tangent line to $z = f(x, y)$ in the yz -plane, an idea we will explore more fully in a later section.

EXAMPLE 3 Find f_x and f_y when

$$f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$$

Solution: To find f_x , we begin with the chain rule

$$f_x = \frac{\partial}{\partial x} \tan^{-1}(\text{input}) = \frac{1}{(\text{input})^2 + 1} \frac{\partial}{\partial x} (\text{input})$$

where the input is y/x . Writing the input as $x^{-1}y$ and substituting then yields

$$f_x = \frac{1}{(x^{-1}y)^2 + 1} \frac{\partial}{\partial x} (x^{-1}y) = \frac{-x^{-2}y}{x^{-2}y^{-2} + 1}$$

To simplify this expression, we multiply the numerator and denominator by x^2 :

$$f_x = \frac{x^2 (-x^{-2}y)}{x^2 (x^{-2}y^{-2} + 1)} = \frac{-x^2 x^{-2} y}{x^2 x^{-2} y^2 + x^2} = \frac{-y}{y^2 + x^2}$$

To find f_y , we again begin with the chain rule,

$$f_y = \frac{\partial}{\partial y} \tan^{-1}(\text{input}) = \frac{1}{(\text{input})^2 + 1} \frac{\partial}{\partial y}(\text{input})$$

where the input is y/x . The result is that

$$f_y = \frac{1}{\left(\frac{y}{x}\right)^2 + 1} \frac{\partial}{\partial y} \left(\frac{y}{x}\right) = \frac{1}{\left(\frac{y}{x}\right)^2 + 1} \left[\frac{1}{x} \frac{\partial}{\partial y}(y) \right]$$

which simplifies to

$$f_y = \frac{1}{\left(\frac{y^2}{x^2} + 1\right) x} = \frac{x}{\left(\frac{y^2}{x^2} + 1\right) x^2} = \frac{x}{x^2 + y^2}$$

Check your Reading: What happened to the expression $x^2 x^{-2}$ in example 3?

Interpretations of the Partial Derivative

Analogous to a function of 2 variables, we define a *function of 3 variables* is a mapping that assigns one and only one real number to each point in a subset of 3 dimensional space. It follows that a function of three variables is of the form

$$F(x, y, z) = \text{“expression in } x, y, \text{ and } z\text{”}$$

Partial derivatives of functions of 3 variables are defined analogously to partial derivatives of functions of two variables (see the exercises). Thus, to differentiate a function of 3 variables $F(x, y, z)$ with respect to x , we differentiate as if y and z are constants. The partial derivatives F_y and F_z are defined similarly, and correspondingly, to calculate F_y and F_z , we differentiate with respect to y and z , respectively.

EXAMPLE 4 Find the first partial derivatives of $F(x, y, z) = x^3 + 3xyz + z^2$.

Solution: To compute f_x , we treat y and z as if they were constant:

$$\begin{aligned} F_x &= \frac{\partial}{\partial x} (x^3 + 3xyz + z^2) \\ &= \frac{\partial}{\partial x} x^3 + 3yz \frac{\partial}{\partial x} x + \frac{\partial}{\partial x} z^2 \\ &= 3x^2 + 3yz \end{aligned}$$

Likewise, F_y follows from treating x and z like constants,

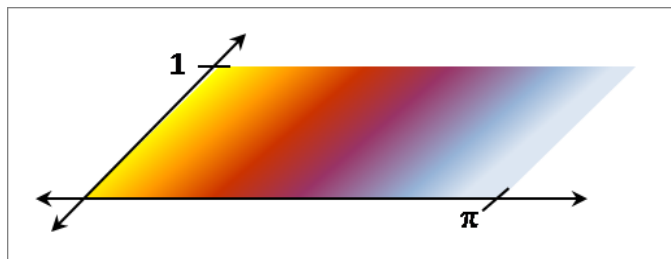
$$F_y = \frac{\partial}{\partial y} (x^3 + 3xyz + z^2) = 0 + 3xz + 0 = 3xz$$

and F_z follows from treating x and y like constants.

$$F_z = \frac{\partial}{\partial z} (x^3 + 3xyz + z^2) = 3xy + 2z$$

In many applications, functions of three variables occur in the form $u(x, y, t)$, where t is a measure of *time*. In such examples, the definition of the partial derivative u_t implies that it is *the rate of change* of $u(x, y, t)$ with respect to t . Indeed, partial derivatives often occur in applications as a rate of change of a given output with respect to only one of several inputs.

EXAMPLE 5 A rectangular sheet of metal with a length of π feet and a width of 1 foot has its left section placed in an oven and its rightmost extent placed in liquid nitrogen.



Upon being removed from the oven and nitrogen, its temperature u in $^{\circ}F$ at time t in seconds and at a point (x, y) on the sheet is given by

$$u(x, y, t) = 75 + 300e^{-0.2t} \cos(x) \cosh(y)$$

How fast is the temperature of the sheet changing with respect to time at the point $(0, 0)$? At the point $(\pi, 0)$? How do the rates compare?

Solution: The rate of change of u with respect to t is given by

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial t} (75 + 300e^{-0.2t} \cos(x) \cosh(y)) = -60e^{-0.2t} \cos(x) \cosh(y)$$

Thus, at $(0, 0)$, the time rate of change of temperature is

$$\frac{\partial u}{\partial t}(0, 0, t) = -60e^{-0.2t} \cos(0) \cosh(0) = -60e^{-0.2t} \text{ } ^\circ F \text{ per sec}$$

while at $(\pi, 0)$, the time rate of change of temperature is

$$\frac{\partial u}{\partial t}(\pi, 0, t) = -60e^{-0.2t} \cos(\pi) \cosh(0) = 60e^{-0.2t} \text{ } ^\circ F \text{ per sec}$$

This means that the temperature increases to $75^\circ F$ over time at the point $(\pi, 0)$, while it decreases down to $75^\circ F$ at $(0, 0)$. **Note:** this fits well with the fact that the temperature initially at $(0, 0)$ is $375^\circ F$, while the temperature initially at $(\pi, 0)$ is $-225^\circ F$.

Check your Reading: After a long period of time, what will be the approximate temperature of the metal sheet at every point on the sheet?

Second Derivatives

The second partial derivative of f with respect to x is denoted f_{xx} and is defined

$$f_{xx}(x, y) = \frac{\partial}{\partial x} f_x(x, y)$$

That is, f_{xx} is the derivative of the first partial derivative f_x . Likewise, the second partial derivative of f with respect to y is denoted f_{yy} and is defined

$$f_{yy}(x, y) = \frac{\partial}{\partial y} f_y(x, y)$$

Finally, the *mixed* partial derivatives are denoted f_{xy} and f_{yx} , respectively, and are defined

$$f_{xy}(x, y) = \frac{\partial}{\partial y} f_x(x, y) \quad \text{and} \quad f_{yx} = \frac{\partial}{\partial x} f_y(x, y)$$

Collectively, f_{xx} , f_{yy} , f_{xy} , and f_{yx} are known as the *second partial derivatives* of $f(x, y)$. Moreover, we sometimes denote the second partial derivatives in the form

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{yx} = \frac{\partial^2 f}{\partial y \partial x}$$

EXAMPLE 6 Find the second partial derivatives of

$$f(x, y) = x^3 + 3x^2y^2$$

Solution: The first partial derivatives are $f_x = 3x^2 + 6xy^2$ and $f_y = 6x^2y$. As a result, we have

$$\begin{aligned}f_{xx}(x, y) &= \frac{\partial}{\partial x} f_x(x, y) = \frac{\partial}{\partial x} (3x^2 + 6xy^2) = 6x + 6y^2 \\f_{yy}(x, y) &= \frac{\partial}{\partial y} f_y(x, y) = \frac{\partial}{\partial y} (6x^2y) = 6x^2 \\f_{xy}(x, y) &= \frac{\partial}{\partial y} f_x(x, y) = \frac{\partial}{\partial y} (3x^2 + 6xy^2) = 12xy \\f_{yx}(x, y) &= \frac{\partial}{\partial x} f_y(x, y) = \frac{\partial}{\partial x} (6x^2y) = 12xy\end{aligned}$$

Notice that the mixed partial derivatives are the same. Indeed, the mixed partials are always the same for “nice” functions, as is stated below in *Clairaut’s theorem*.

Clairaut’s Theorem: If f is defined on a neighborhood of (p, q) and if f_{xy} and f_{yx} are continuous on that neighborhood, then

$$f_{xy}(p, q) = f_{yx}(p, q)$$

There are functions that do not satisfy the hypotheses of theorem 3.1 for which the mixed partials are not the same at some point (see exercise 46). However, our focus will be on functions with continuous second partial derivatives, in which case the mixed partials are the same (a proof of Clairaut’s theorem is given in chapter 4).

EXAMPLE 7 Find f_{yx} and f_{xy} for $f(x, y) = x \sin(xy)$

Solution: The first partial derivatives are

$$f_x = \sin(xy) + xy \cos(xy) \quad \text{and} \quad f_y = x^2 \cos(xy)$$

The product rule thus implies that

$$f_{yx} = \frac{\partial}{\partial x} (x^2 \cos(xy)) = 2x \cos(xy) - x^2 y \sin(xy)$$

Now let's compute f_{xy} (and thus confirm theorem 3.1):

$$\begin{aligned} f_{xy} &= \frac{\partial}{\partial y} (\sin(xy) + xy \cos(xy)) \\ &= x \cos(xy) + x \cos(xy) + xy \left(-\sin(xy) \frac{\partial}{\partial y} xy \right) \\ &= 2x \cos(xy) - x^2 y \sin(xy) \end{aligned}$$

Check your Reading: If $f(x, y, z)$ is infinitely differentiable in each variable, then is $f_{xz} = f_{zx}$?

Higher Derivatives

Higher partial derivatives are defined similarly. For example, the third derivative of f with respect to x is the partial derivative with respect to x of the second derivative f_{xx} . That is,

$$f_{xxx}(x, y) = \frac{\partial}{\partial x} f_{xx}(x, y)$$

Similarly, f_{xxy} is defined

$$f_{xxy}(x, y) = \frac{\partial}{\partial y} f_{xx}(x, y)$$

In operator notation, the partial derivative of f for m times with respect to x and n times with respect to y is denoted by

$$\frac{\partial^{m+n} f}{\partial x^m \partial y^n}$$

The $m+n$ partial derivatives of $f(x, y)$ are then defined in terms of the previous partial derivatives as

$$\frac{\partial^{m+n} f}{\partial x^m \partial y^n} = \frac{\partial}{\partial x} \frac{\partial}{\partial y} \left(\frac{\partial^{m+n-2} f}{\partial x^{m-1} \partial y^{n-1}} \right)$$

when $f(x, y)$ and its partial derivatives are continuous on a region through the $m+n$ order.

EXAMPLE 8 Find f_{xxyy} if $f(x, y) = x^4y^4$.

Solution: It is easy to show that $f_{xx} = 12x^2y^4$. Thus,

$$f_{xxy} = \frac{\partial}{\partial y} f_{xx} = \frac{\partial}{\partial y} 12x^2y^4 = 48x^2y^3$$

and similarly,

$$f_{xxyy} = \frac{\partial}{\partial y} f_{xxy} = \frac{\partial}{\partial y} 48x^2y^3 = 144x^2y^2$$

Moreover, we usually assume that f is sufficiently smooth at all points where partial derivatives are defined so that mixed partials are independent of the order of differentiation. Indeed, notice that if $f(x, y) = x^4y^4$, then

$$f_{xyx} = \frac{\partial}{\partial x} f_{xy} = \frac{\partial}{\partial x} 16x^3y^3 = 48x^2y^3$$

which is the same as f_{xxy} in example 8. In addition, $f_{xyxy} = 144x^2y^2 = f_{xxyy}$.

Exercises

Find $f_x(x, y)$ and $f_y(x, y)$ for each of the following:

- | | |
|-------------------------------------|---------------------------------------|
| 1. $f(x, y) = x^2 + y^3$ | 2. $f(x, y) = x^2 + 2xy + y^3$ |
| 3. $f(x, y) = (x + 2y)^2$ | 4. $f(x, y) = (x^2 + 2y)^2$ |
| 5. $f(x, y) = x \sin(y)$ | 6. $f(x, y) = e^x \ln(y^2 + 1)$ |
| 7. $f(x, y) = \exp(-x^2 - y^2)$ | 8. $f(x, y) = \tan^{-1}(xy)$ |
| 9. $f(x, y) = x \cos(xy)$ | 10. $f(x, y) = x \sin(xy)$ |
| 11. $f(x, y) = y^x$ | 12. $f(x, y) = x^y$ |
| 13. $f(x, y) = \frac{x}{x^2 + y^2}$ | 14. $f(x, y) = \int_x^y \sin(t^2) dt$ |

Find f_{xx} , f_{xy} , f_{yx} , and f_{yy} for each of the following. Then show that the mixed partials are the same.

- | | |
|----------------------------|----------------------------------|
| 15. $f(x, y) = x^2 + y^3$ | 16. $f(x, y) = x^2 + 2xy + y^3$ |
| 17. $f(x, y) = (x + 2y)^2$ | 18. $f(x, y) = (x^2 + 2y)^2$ |
| 19. $f(x, y) = x \sin(y)$ | 20. $f(x, y) = e^x \ln(y^2 + 1)$ |
| 21. $f(x, y) = x \cos(xy)$ | 22. $f(x, y) = \tan^{-1}(xy)$ |
| 23. $f(x, y) = y^x$ | 24. $f(x, y) = x^y$ |

Find the indicated derivative of the given function:

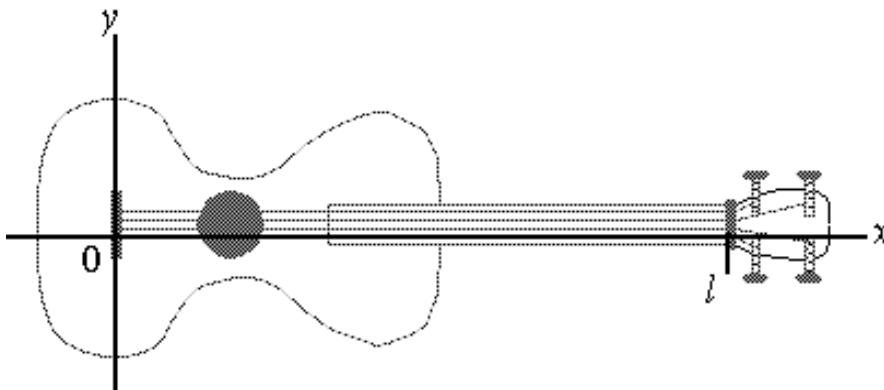
- | | |
|--|--|
| 25. f_{xxy} for $f(x, y) = x^2 + y^3$ | 26. f_{xxy} for $f(x, y) = x^2 + 2xy + y^3$ |
| 27. f_{xyx} for $f(x, y) = (x + 2y)^2$ | 28. f_{yxy} for $f(x, y) = (x^2 + 2y)^2$ |
| 29. f_{xxyy} for $f(x, y) = x \sin(y)$ | 30. $f_{xxxxxxy}$ for $f(x, y) = e^x \ln(y^2 + 1)$ |
| 31. f_{xxxxy} for $f(x, y) = y \cos(xy)$ | 32. f_{xyy} for $f(x, y) = \sin(x) \tan(xy)$ |

33. A vibrating string has a displacement $y = u(x, t)$ in cm at a distance x in cm from one end and at time t in seconds, where

$$u(x, t) = 2 \sin(120\pi(x - t))$$

How fast (in units of cm per sec) is the string vibrating at a horizontal distance $x = 1.2$ cm from one end at time $t = 2$ seconds? At time $t = 3$ seconds?

34. Suppose that a string is attached at its endpoints $x = 0$ and $x = l$, for some number l .



Suppose also that $y = u(x, t)$ models the displacement at x in $[0, l]$ of the string at time t , where

$$u(x, t) = A \cos(at) \sin\left(\frac{\pi}{l}x\right)$$

with A and a constant.

1. (a) Show that $u(0, t) = u(l, t) = 0$ for all t . How does this relate to $u(x, t)$ being a model of a string?
- (b) What is the rate of change of $u(x, t)$ at $x = l/2$ at any given time?

35. The function $u(x, t) = e^{-t} \sin^2(\pi x) + 32$ models the temperature in $^{\circ}F$ of a 1 foot long thin rod in which both ends are held at the freezing point at all times t . How fast is the temperature decreasing at the midpoint of the rod when $t = 0$? When $t = 1$? When $t = 2$?

36. The function $u(x, y, t) = 2 \sin(3x) \sin(4y) \cos(5t)$ models the displacement u in cm of a vibrating rectangular membrane at time t in seconds and at

a point (x, y) on the membrane. How fast is the displacement of the membrane above the point $(1, 1)$ changing with respect to time at $t = 1$ seconds?

37. It can be shown that an ideal gas with fixed mass has an absolute temperature R , a pressure P , and a volume V that satisfies

$$T = kPV$$

where k is a constant. How fast does the temperature T change with respect to the volume V ?

38. The total resistance R produced by two resistors with resistances R_1 and R_2 , respectively, satisfies

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

What is the rate of change of the total resistance R with respect to the resistance R_1 ?

39. If two planets with masses M and m are located at the points (x, y, z) and $(0, 0, 0)$, respectively, then the potential energy of their mutual gravitational attraction is given by

$$\phi(x, y, z) = G \frac{Mm}{\sqrt{x^2 + y^2 + z^2}}$$

where G is the universal gravitational constant. At what rate is the potential energy changing with respect to x ? With respect to y ?

40. A *Cobb-Douglas* production function is a function of the form $P = bL^\alpha K^\beta$ where b, α , and β are constants. What is the rate of change of P with respect to L ? With respect to P ?

41. Suppose we consider $f(x, y) = x^2 + y^2$.

1. (a) What is the slope of $z = f(x, y)$ for $(p, q) = (1, 2)$ in the x -direction? in the y -direction?
- (b) What curve is formed by the intersection of the plane $y = 2$ with the surface $z = x^2 + y^2$? How does it relate to $f_x(1, 2)$?
- (c) What curve is formed by the intersection of the plane $x = 1$ with the surface $z = x^2 + y^2$? How does it relate to $f_y(1, 2)$?

42. Suppose we consider $f(x, y) = x^2 + xy$.

1. (a) What is the slope of $z = f(x, y)$ for $(p, q) = (1, 2)$ in the x -direction? in the y -direction?
- (b) What curve is formed by the intersection of the plane $y = 2$ with the surface $z = x^2 + y^2$? How does it relate to $f_x(1, 2)$?
- (c) What curve is formed by the intersection of the plane $x = 1$ with the surface $z = x^2 + y^2$? How does it relate to $f_y(1, 2)$?

43. If $f(x, y) = g(x) + h(y)$, then what is f_x and f_y ? What is the equation in x and z of the curve formed by the intersection of $z = f(x, y)$ with the vertical plane $y = q$? with the vertical plane $x = p$? How are these curves related to $f_x(p, q)$ and $f_y(p, q)$, respectively?

44. If $f(x, y) = g(x) + h(y)$, then what is f_x and f_y ? What is the equation in x and z of the curve formed by the intersection of $z = f(x, y)$ with the vertical plane $y = q$? with the vertical plane $x = p$? How are these curves related to $f_x(p, q)$ and $f_y(p, q)$, respectively?

45. If f_x and f_y both exist, how can the limit

$$\lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y+h)}{h}$$

be expressed in terms of the 1st partial derivatives of f ?

46. Write to Learn: Write a short essay in which you use the following steps to show that

$$f(x, y) = \begin{cases} \frac{x^2y - y^2x}{x+y} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$, that $f_x(x, y)$ and $f_y(x, y)$ are continuous at $(0, 0)$, but that

$$f_{xy}(0, 0) \neq f_{yx}(0, 0)$$

1. (a) Show that if $(x, y) \neq (0, 0)$, then

$$f(x, y) = \frac{x - y}{\frac{1}{x} + \frac{1}{y}}$$

and correspondingly

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} \frac{x - y}{\frac{1}{x} + \frac{1}{y}} = 0$$

- (b) Show that if $(x, y) \neq (0, 0)$, then

$$f_x(x, y) = \frac{y(x^2 + 2xy - y^2)}{(x + y)^2}$$

and explain why that as a result, we have

$$\lim_{(x, y) \rightarrow (0, 0)} f_x(x, y) = 0$$

- (c) Define $f_x(0, 0) = 0$ and then evaluate

$$f_{xy}(0, 0) = \lim_{h \rightarrow 0} \frac{f_x(0, 0+h) - f_x(0, 0)}{h}$$

- (d) Repeat (b) and (c) beginning with the fact that

$$f_y(x, y) = \frac{x(x^2 - 2xy - y^2)}{(x + y)^2}$$

The outcome should be that $f_{yx}(0, 0)$ is not the same as $f_{xy}(0, 0)$.